The modified step works; the printed rule is unresolved

Fixed-penalty Quiroga-style FGD • audited Ruche experiments • two-qubit diagnostic

Exact factor-ray line search recovers the target under the unchanged physical gate. It changes the adaptive step, not the objective or gradient direction. This is a successful explicit variant—not a reproduction of published adaFGD.

What was tested

Two qubits; rank-one Haar-unitary channels; 128 noiseless Born observations; normalized Gell-Mann initialization. Same data and starting factor within each channel. Gate: fidelity ≥ 0.99, full-design RMSE ≤ 1e-6, normalized TP violation ≤ 1e-3. No projection or relaxed gate.

New screen: 50 runs on calibration channels 0/1. Literal rule: four fixed penalties × six constant step multipliers, excluding two previously archived settings per channel. Exact-ray variant: three fixed penalties per channel; maximum 10,000 updates for every run.

Screen result: literal and modified methods remain separate

Printed adaptive ratio	0/44 pass: 14 completed, 30 numerical failures
Exact factor-ray variant	6/6 pass and converge; all three penalties succeed on both channels

Frozen choice: $\alpha = 5e\text{-}5$, $\lambda = 128\alpha = 0.0064$, fixed throughout each trial. Selection minimized the worst-channel normalized gate deficit; it did not select the fastest wall time. Selected screen RMSE: 8.93e-11–1.00e-10; updates: 1401–1882.

Frozen follow-up evidence

Stage / channels	Gate	Endpoint RMSE range	Updates
Selected screen / 0–1	2/2	8.93e-11–1.00e-10	1401–1882
Calibration confirmation / 2–7	6/6	7.73e-11–1.60e-10	794–1418
Historical revalidation / 100–107	8/8	4.83e-11–1.61e-10	816–1790

Frozen setting: 16/16 gates pass; 16/16 converge. Minimum fidelity 1 (rounded); maximum normalized TP 7.80e-10.

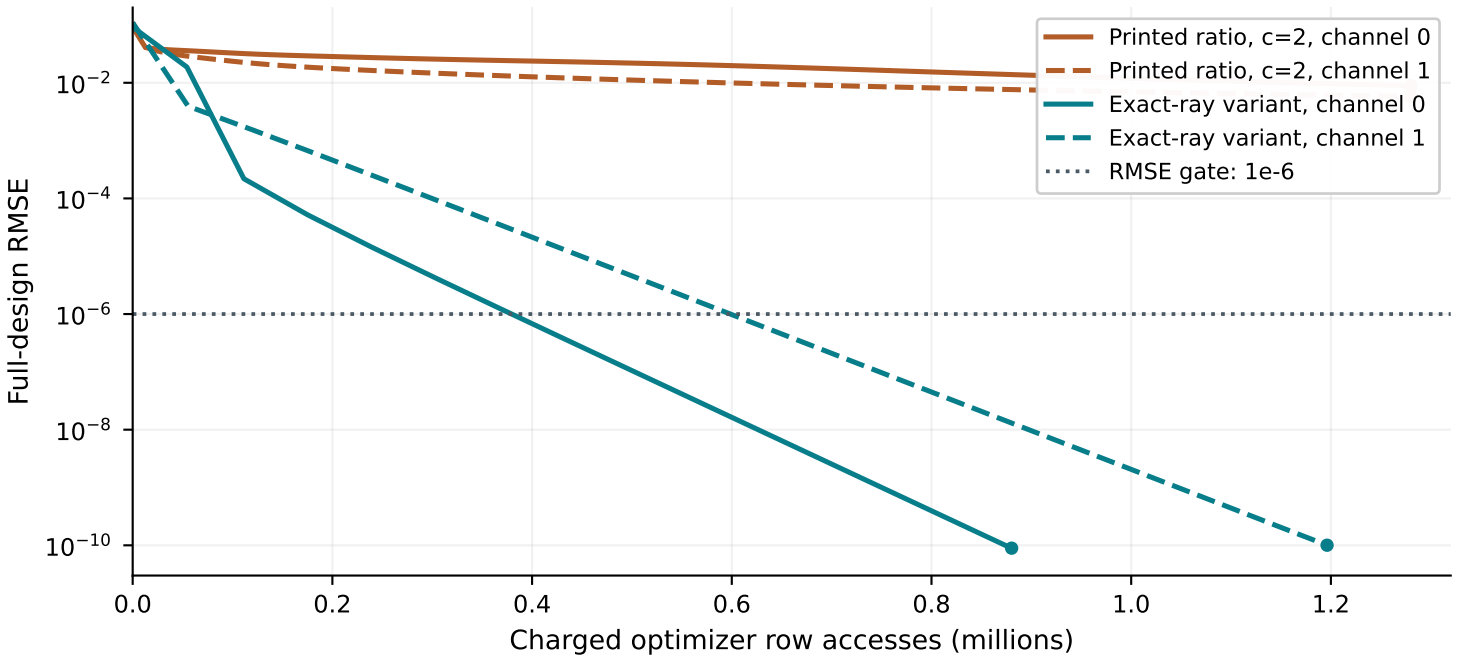
Channels 2–7 are additional calibration. The unchanged setting passes all historical channels 100–107; this is not unseen validation. No retuning; held-out channels 1000–1019 remain untouched.

Conclusion and remaining boundary

The fixed-penalty factorized formulation is recoverable here; the tested printed ratio is the unresolved part. This is not evidence that Quiroga's paper is wrong. No completed higher-qubit campaign or final best-for-best FRAMES comparison follows from these tests.

Why the step-size rule is the focus

Same α = 5e-5 and λ = 0.0064 • paired calibration channels 0/1 • no optimizer reruns



Iterations do not have equal cost. Exact-ray accounting includes the direction pass, two polynomial forward passes and each directly evaluated candidate. Charges exclude TP contractions, spectral-norm work, full-design overhead and checkpoint evaluation; this is not a wall-time or total-compute equivalence.

Why no blind 100,000-update extension was selected

Best jointly stable literal setting: α = 5e-5, c = 2; worst RMSE 0.0089485, still 8,948× the gate after 10,000 updates. Recorded tail slopes imply only ~7e-4 and ~1.1e-3 at 100,000 updates if those rates persisted. These are diagnostic extrapolations, not guarantees; they do not justify the optional extension.

Prior context, not new success: four GM-initialized printed-rule runs already reached fidelity > 0.99 at 100,000 updates, but all failed the joint RMSE/TP gate. All earlier results and all numerical failures remain preserved.

Checks and provenance

Source 4ce1108a0692; full suite: 597 passed, 55 skipped; tuning harness: 16 passed. Audits verify pins, archives, independent endpoints, gates, initialization, stopping and work. Screen Slurm exit 2 denotes archived method failures, not missing results. Ruche jobs: screen 2039775; confirmation 2039794; revalidation 2039814.

Relation to the published work

Quiroga and Kyrillidis give an adaptive ratio only up to proportionality. The multiplier, experimental TP weight and stopping choices are not fully specified for exact reproduction. Their cited reference [47] treats hard-thresholding matrix methods. Our positive-factor-ray quartic line search is neither the printed adaFGD step nor Matrix ALPS.

Quiroga & Kyrillidis, arXiv:2312.01311, §III-B

[47] Kyrillidis & Cevher, arXiv:1203.4481

Screen audit SHA256: ce5135b4e04535bf90fa143cf7372341428fb15217986c8a8443f5536f356074